

Table of Contents

TOP Systematic Uncertainties (Run2).....	1
Luminosity.....	1
Pile up.....	1
Trigger efficiency.....	2
Muon identification, isolation, momentum scale and resolution.....	2
Electron and photon identification.....	3
Jets and MET.....	3
b tagging.....	4
Forward protons.....	4
Modelling uncertainties in general.....	5
PDF.....	5
Comparison with other event generators.....	5
Factorization and renormalization scales at matrix-element level.....	6
Parton shower uncertainties.....	6
Parton shower: ISR and FSR governed by S at scale Q^2	6
Parton shower: ME-PS matching scale (hdamp).....	7
Parton shower: Color Reconnection.....	7
Parton shower: b-fragmentation.....	8
---+++ Parton shower: Decay tables.....	8
Underlying event.....	9
Top pt reweighting.....	9
Mass of the top quark.....	9
Recommendations for single top analyses.....	11

TOP Systematic Uncertainties (Run2)

General prescriptions for the evaluation of the main sources of systematic uncertainties in TOP analyses are given below. Along with the recipes and detailed recommendations we include when available the public documents that should be referenced in the documentation of the analyses.

Before you start

- The list of uncertainties below is not meant to be exhaustive or relevant for every analysis. Consider the implementation of each uncertainty carefully!
 - ◆ There are specific experimental effects that can affect only **your** analysis. Start with these [slides](#) as an inspiring introduction.
 - ◆ Complimentary measurements can reduce the impact of these uncertainties. Try as much as possible to come up with **ancillary/in-situ** measurements.
 - ◆ Some recommendation for likelihood-based analyses with **profiled systematics** can be found [here](#).
 - ◆ Analysis performing **unfolding** of distributions can check these slides [for](#) available methods and helpful hints.

Don't hesitate to contact the TOP PAG conveners and the TOP PAG contacts for clarifications or to propose corrections. The corresponding page for Run1 is TopSystematicsRun1.

Instructions for the conventional naming of systematic uncertainties (important for all analyses using the Higgs Combine Tool) are linked on the TopCombine page.

Luminosity

Last check: Feb. 2024

Please follow the Lumi POG recommendations, documented and maintained here:
<https://twiki.cern.ch/twiki/bin/view/CMS/TWikiLUM>

The relevant papers to cite can also be found on that Twiki page.

Shortcuts:

<https://twiki.cern.ch/twiki/bin/view/CMS/LumiRecommendationsRun2>

<https://twiki.cern.ch/twiki/bin/view/CMS/LumiRecommendationsRun3>

Run 2 Combination and correlations

In case the luminosity gets updated after publication, following a PC decision in 2021 we usually do not fold in the updates, e.g. in errata, in particular if it requires performing a fit again. However, it may and should be updated if it is easy for summary plots or if the result is used subsequently for a parameter extraction (and the luminosity matters). For review papers, no update is needed in most cases.

Pile up

Last check: Feb. 2024

Most relevant link: https://twiki.cern.ch/twiki/bin/view/CMS/TWikiLUM#Pileup_information

Correlation: Fully correlated across the years for the uncertainty of 4.6% (Ref.). The part of the uncertainty that covers the analysis level difference of data and MC should be treated as correlated (uncorrelated) if the discrepancy is found to have consistent (inconsistent) features among the years.

Trigger efficiency

Last check: Feb. 2024

For details about Trigger efficiencies, please see TopTrigger.

- Correlations: when using the orthogonal method to measure trigger efficiencies, and as long as scale factors are derived year-by-year, the uncertainties should be considered uncorrelated across the years. Other recommendations may apply when using triggers derived with other methods or by the POGs.
- The scale factors derived with data are available in Top trigger scale factors.
- Note: new combination of trigger paths as well as new methods for deriving trigger scale factors must be approved by relevant groups, in coordination with the TOP Trigger contacts.

Feel free to interact with our Top trigger experts if you need help with your trigger scale factors and/or assigning proper uncertainties, or inform them of your estimations so that they can be useful to other analysts in similar final states.

Muon identification, isolation, momentum scale and resolution

Last check: Feb. 2024

For details about Muon identification, please see TopMUO.

- Correlations: The following correlation scheme should be applied to the uncertainties:
 - ◆ Systematic uncertainties on the scale factor measurement in Z events should be considered correlated among years.
 - ◆ Statistical uncertainties on the scale factor measurement should be considered uncorrelated among the different years.
 - ◆ Extrapolation uncertainties to top quark events should be considered correlated among years.
 - ◆ Uncertainties are mostly uncorrelated between electrons and muons. Exceptions can happen in cases like the additional extrapolation uncertainty where, depending on the method, strong correlations can be induced. Please get in touch with TOP MUO contacts for those cases.
 - ◆ An analysis using 2016, 2017 and 2018 data should have 5 separate sources of systematic uncertainty from the muon ID efficiency
 - ◇ 1 systematic on SF measurement, 3 statistical on SF measurement, i.e., one each year, and 1 for extrapolation
 - ◇ Could be even more if separate ID and ISO scale factors are applied.
- **Note:** new methods for deriving scale factors or for the identification, isolation and/or momentum scale and resolution must be approved by the MUO POG, in coordination with TOP MUO contacts.

Extrapolation to top topologies. Note that the scale factors provided by the POG are computed on a Z sample which can differ from scale factors from top events. This can potentially be a concern for the lepton isolation scale factors as the top events tend to have topologies with more jets. Such effects have been studied for electrons and muons and documented in the following references

- this note [using in-situ methods](#)
- this ATLAS publication [using in-situ methods](#)
- this presentation [, using DY events in phase spaces with large multiplicity](#)

The latter method has been studied with more detail and more recently, therefore it is the recommended approach, leading to uncertainties at the permille level. With that method, the extrapolation from the DY to the ttbar phase space is taken into account as an additional multiplicative correction with its associated corrections. correctionlib json files can be found [here](#).

In the combine fit treat this set of scale factors (describing the isolation extrapolation to ttbar) as a separate nuisance parameter in the datacard from the central identification/isolation scale factors. These SFs are correlated between years.

Electron and photon identification

Last check: Feb. 2024

For details about Electron and Photon reconstruction and identification, please see TopEGM.

- Correlations: the EGamma POG recommendation on combining the systematic uncertainties among 2016-2018 years can be found [here](#).
- **Note:** new methods for identification and deriving scale factors has to be approved by the EGamma POG, in coordination with TOP EGM contacts.

Extrapolation to top topologies. Note that the scale factors provided by the POG are computed on a Z sample which can differ from scale factors from top events. This can potentially be a concern for the lepton isolation scale factors as the top events tend to have topologies with more jets. Such effects have been studied for electrons and muons and documented in the following references

- this note [using in-situ methods](#)
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In the combine fit treat this set of scale factors (describing the isolation extrapolation to ttbar) as a separate nuisance parameter in the datacard from the central identification/isolation scale factors. These SFs are correlated between years.

Jets and MET

Last check: Feb. 2024

Recommendations by JME:

https://twiki.cern.ch/twiki/bin/view/CMS/JetMET#Quick_links_to_current_recommend

For details on jet energy scales, resolutions, and MET reconstruction specific to top analyses see TopJME.

- **Jet energy scale**

- ◆ For the evaluation of jet energy scale uncertainties use the recipe documented here.
- ◆ Correlation and grouping for Run 2

- ◇ Analyses sensitive to JECs: Table for all sources [\[2\]](#) [\[3\]](#). Valid across all three years. **All analyses are required to evaluate the whole set.**
- ◇ Analyses mildly sensitive to JECs: Table for reduced sources [\[2\]](#) [\[3\]](#) (prototype). A description for reduced sources is provided here. Sources across the years are treated as 100% correlated. **This reduced set is not recommended in TOP and should only be used in special cases after discussion with conveners.**

- **Jet energy resolution**

- ◆ Correlation: according to this page, 6 nuisance parameters must be assigned per year. It is recommended by JETMET to treat JER as uncorrelated across all years. Show uncertainty splitting
- **Flavor uncertainty:** TOP analyses are sensitive to the different jet responses for gluon, light quarks, charm and b quarks. JME provides JEC uncertainty for each pure jet flavor separately, This breakdown should be used as a default and will generally reduce the total uncertainty.
- - ◆ Correlation across the years: this uncertainty is considered correlated for the same process across the years.
 - ◆ Correlation between singletop and ttbar: this uncertainty is considered correlated among all processes.

b tagging

Last check: Feb. 2024

Scale factor recommendations by BTV: <https://btv-wiki.docs.cern.ch/ScaleFactors/>

[\[2\]](#)Uncertainties and correlation recommendations by BTV:
<https://btv-wiki.docs.cern.ch/PerformanceCalibration/SFUncertaintiesAndCorrelations/>

[\[2\]](#)SF procedures:

- Use fixedWP SFs wherever possible, use iterative fit SFs only if the b-tagging discriminant shape is used somewhere in your analysis
- Recommendations for working point based b-tagging scale factors (fixedWP):
<https://btv-wiki.docs.cern.ch/PerformanceCalibration/fixedWPSFRecommendations/>
 - ◆ Note for TOP analyses: The b-jet SF combination `comb` contains SF derivation methods from ttbar-enriched phase spaces. Please check the overlap with those before using the `comb` SFs, and use the `mujets` SFs if there is any notable overlap.
 - ◆ The same holds for analyses sensitive to V_{tb} , which have to use `mujets` SFs.
- Recommendations for b-tagging shape reweighting scale factors (iterative fit):
<https://btv-wiki.docs.cern.ch/PerformanceCalibration/shapeCorrectionSFRecommendations/> [\[2\]](#)

Citation recommendations by BTV: <https://btv-wiki.docs.cern.ch/PerformanceCalibration/Citations/> [\[2\]](#)

Forward protons

Last update: April 2021

- **Proton scale factors:** The recommended proton efficiency scale factors and the associated uncertainties are provided here
- **Uncertainties.** Besides the SF uncertainties, an additional flat uncertainty should be applied for pixels ([link](#)) and strips ([link](#)).
- **Correlation:** Uncertainties are treated uncorrelated between the years.
- More information can be found in TaggedProtonsPOGRecommendations.

Modelling uncertainties in general

Last check Feb. 2024

In general, modeling uncertainties are correlated across all years.

Pre-acceptance normalization: In general modeling uncertainties that change the total cross section should be normalized back to the reference cross section before any acceptance requirements.

In case of differential measurement this should be done per bin at the relevant level (parton/particle).

For nuisance parameter fits, It is generally encouraged to apply constraints on the modeling uncertainties only in the visible phase space. For the extrapolation to the full phase space, these uncertainties should be added (again) for the acceptance correction when extrapolating to the full phase space

PDF

Last check Feb. 2024

- As the minimal starting point, use the set NNPDF31_nnlo_as_0118_mc_hessian_pdfas (lhaid: 325300) including Hessian eigenvectors and the two α_s variations in the event weights available in our simulated samples, see e.g. MC PDF Weights and Scales
- Please use one PDF set (e.g. NNPDF3.1 used as default in the MC). By default consider each eigenvector as an independent nuisance parameter and the variation of the strong coupling constant as an additional one. These eigenvectors should be treated as correlated (one-by-one) across processes in fits.
- Correlations: each uncertainty vector is considered fully correlated across the years and between processes.

Comparison with other event generators

Last check Feb. 2024

We don't quote the difference between event generators as an uncertainty, with event generators being Powheg vs MG (LO) or aMC@NLO. However, we encourage the analyses to test all available generators and document in the respective AN the observed effect (e.g. by looking at FxFx if available). This important information is pertinent to subsequent comparisons with other experiments. Differential measurements are expected to have these comparisons included in the distributions.

For ttbar modelling we have in production Powheg and aMC@NLO samples which are both NLO in precision, being the so-called aMC@NLO FxFx sample precise at NLO up to two extra jets (and LO after the third jet emission). As reported in TOP-16-021 [\[1\]](#) [\[2\]](#) the matching to the parton-shower needs more attention from the MC developers for aMC@NLO FxFx. Thus, a direct comparison of Powheg vs aMC@NLO with the current settings is expected to lead to inflated differences in systematics which are not fully understood and can't yet be fully translated into a physically meaningful systematic uncertainty

Please also see the section Modeling uncertainties in general above

Factorization and renormalization scales at matrix-element level

Last check Feb. 2024

- We recommend quoting the following **separate variations** (nuisance parameters) for the QCD scale from ME weights QCD scale from ME weights
 - ◆ factorization scale: μ_R fixed, vary μ_F by factor 2 (0.5) for up (down) variation
 - ◆ renormalization scale: μ_F fixed, vary μ_R by factor 2 (0.5) for up (down) variation
- Please note that it is important for a template fit that the factorization and renormalization scale uncertainties as separate nuisance parameters, instead of constructing an envelope.
- Correlation across the years: this uncertainty is considered correlated for the same process across the years.
- Correlation between $t\bar{t}$ and single-top: this uncertainty is considered uncorrelated between QCD-induced ($t\bar{t}$) and EWK-induced (single-top) processes. See e.g. TOP-17-001[\[4\]](#).

Please also see the section [Modeling uncertainties in general](#) above

Parton shower uncertainties

Parton shower: *ISR and FSR governed by α_s at scale Q^2*

Last check: Feb 2024

The assumed value of the strong coupling constant α_s at scale Q^2 in the parton shower has a corresponding uncertainty. The uncertainty is evaluated by varying the renormalization scale for QCD emissions in initial-state and final-state radiation (ISR and FSR), effectively varying the value of α_s .

- The uncertainty is evaluated via per-event weights. The weights corresponding to three sets of variations, separately for ISR and FSR, are present in all recent Run-2 MC samples:
 - ◆ Reduced variations: factor $\sqrt{2}$ and $1/\sqrt{2}$
 - ◆ Default variations: factor 2 and 0.5
 - ◆ Conservative variations: factor 4 and 0.25
- Recommendation: default variations with factor 2 and 0.5
- These variations contain NLO compensation terms that preserve the soft-gluon limit[\[4\]](#).
- The variations in Pythia are specified in this page[\[4\]](#)
- A further level of detail: 32 decorrelated variations of the renormalization scale and non-singular terms (cNS) for each branching type (g gg, g qq, q qg, Q Qg with $Q = t$ or b), for ISR and FSR separately
 - ◆ Non-singular terms important to estimate the sensitivity to missing higher-order corrections in the matrix element
 - ◆ See talk by M. Seidel [\[4\]](#) for more information.
 - ◆ Recommended to assess the individual 32 variations if any of the following is true:
 - ◇ PS uncertainty dominant in analysis
 - ◇ nuisance parameter for total variation strongly constrained
 - ◇ similar amount of QCD-induced ($t\bar{t}$, tV) and EWK-induced (single-top, tV) events
 - ◆ Note that these 32 variations are only present in samples produced after (approximately) September 2018.
- The numbering scheme for PS scale variations in the MC samples is detailed here.

- Correlation across the years: this uncertainty is considered correlated for the same process across the years.
- Correlation between singletop and ttbar: or in general, QCD-induced vs EWK-induced processes
- **ISR**: it is advised to not correlate ttbar and single top for ISR. The PS approximates some higher-order ME, and those are process-dependent. For example ME of additional gluon emission in single top is different than that one in ttbar. Even if the full ME for one process is evaluated, it is not known what the true emission rate for another process is. In other words, PS is a relatively crude approximation so if one tunes it to one process and approximate somehow the true ME of that process, it doesn't help for other processes. *Note*: Please note that using the so-called (32) decorrelated PS weights does not justify the correlation between ttbar and single-top given the reasons above. The goal of the decorrelated PS weights is mostly to decorrelate the different branching types (q qg, g gg, ...) within the same process.
- **FSR**: correlated unless the analysis is too sensitive to off-shell effects.

Parton shower: *ME-PS matching scale* (h_{damp})

Last check: Feb 2024

- The POWHEG generator scales the cross section for real emissions by a damping function $hdamp^2 / (pT^2 + hdamp^2)$, which controls the ME-PS merging and regulates the high-pT radiation. The damping variable $hdamp$ is set to 1.379 times m_t in Run 2, a value derived from data at $\sqrt{s} = 8$ TeV in the dilepton channel using a similar ME calculation and assuming the CP5 tune. This value of $hdamp$ was obtained by tuning the distribution of the leading additional jet pT. For Run 3 $hdamp$ was increased to 250 GeV to synchronize the setting with ATLAS as part of the common Powheg hvq effort.
- The dedicated samples with $hdamp$ up and down correspond to the value of $hdamp$ varied up or down according to the uncertainty on the nominally tuned $hdamp$ parameter. For the CP5 tune $hdamp = (1.379 + 0.926 - 0.5052) m_t$ in Run 2, $hdamp = (1.449 + 0.9739 - 0.5333) m_t$ in Run 3. More information can be found in Eur. Phys. J. C 80 (2020) 4 [\[1\]](#) (GEN-17-001), [here](#) [\[2\]](#) and this talk for Run 3 $hdamp$ setting [\[3\]](#).
- Correlation across the years: this uncertainty is considered correlated for the same process across the years.
- Correlation between single top and ttbar: or in general, QCD-induced vs EWK-induced processes for single top one should not assess the ME-PS matching scale using the $hdamp$ variations at this moment since it is not clear whether these make sense (see slide 3 of this talk [\[4\]](#)). Additional studies are required. Please volunteer to contribute to these studies!
- ML reweighting (NEW): to reweight the nominal sample with the CMS default value of $hdamp$ to the CMS up and down variations of $hdamp$, you can follow the instructions in the ML reweighting twiki.

Parton shower: *Color Reconnection*

Last update: Feb 2024

- The differences between the nominal CR model, which is MPI based, and other models should be assessed. These models are:
 - ◆ QCD-inspired (CR1): [link to python script](#) [\[1\]](#)
 - ◆ Gluon move (CR2): [link to python script](#) [\[2\]](#)
 - ◆ MPI based tune with early resonance decays on (ERDon), PartonLevel:earlyResDec = on
- More details on the description of these models can be found in GEN-17-002 [\[3\]](#) and in this presentation [\[4\]](#) or references therein.

- In case you have systematics including CR as nuisance in the fit, each CR variation should enter the fit separately (no envelop).
- Correlation across the years: this uncertainty is considered correlated for the same process across the years.
- Correlation between singletop and ttbar: this uncertainty is correlated between QCD-induced (ttbar, ttV) and EWK-induced (single-top, tV).
- The correlation between the CR tunes and the UE tunes uncertainty is being discussed since in the CR tunes the UE parameters are tuned simultaneously with the CR model parameters.

Parton shower: *b*-fragmentation

Last update: Feb 2024

- Please note that the currently implemented weights to derive the 1SD uncertainty via bowler and perterson variations are causing unphysically large uncertainties of 50-100% - please check slides in TOP PAG agenda for details. We currently do not advise to implement these but handle discussion during pre-app on how it can be handled specifically to the analysis.
- We denote by fragmentation the momentum transfer from the b quark to the B hadron. This can be tuned using LEP/SLD data and the result of the tune can be found in this page [\[4\]](#). The tuned value for the Bowler-Lund parameter, assuming the CP5 tune, is
 $\text{StringZ:rFactB} = 1.056$
 $+0.196-0.200$ which encloses the default used in the CP5 tune, i.e., 0.855. In order to avoid generating extra samples with variations of Bowler-Lund parameter, it was found in Run I top mass analyses that a reweighting at generator level of the transfer function
 $x_b = p_T(\text{B hadron})/p_T(\text{b-jet})$
 is enough to cover most of the effects on the kinematics/response of the b-jets. Thus this uncertainty is proposed to be evaluated based on the generator level value of x_b .
- The weights necessary to bring CP5 to the central value of the tune or to the ± 1 variations and an example code of how to apply them (or to make them available in the EDM event via value maps to be applied to gen-jets produced by the ParticleLevelProducer) can be found [gitlab \[4\]](#). Several weights are made available, but the basic recommendation is to only use the central, up and down variations pt-dependent weights for the Bowler-Lund model (fragCP5BL/fragCP5BLdown/fragCP5BLup). In addition, a comparison with the Peterson model (fragCP5Peterson) can be made. Note that the b jets used in that procedure are genJets with neutrinos included in the clustering, contrary to regular genJets. The weights computed for every b jet in the event should be multiplied to yield an event-level weight.
- Correlation across the years: this uncertainty is considered correlated for the same process across the years.
- Correlation between singletop and ttbar: this uncertainty is considered correlated among QCD-induced (ttbar, ttV) and EWK-induced (single-top, tV) processes.
- Note Blindly applying the weights can lead to biases and effects that are not due to the b fragmentation but to neglected correlations of the weights with other quantities (see the discussion in these slides, 15-17 [\[4\]](#)). This bias should be measured and removed by the analyzers. If the effect seen after correction remains too strong, the weights cannot be trusted to reliably estimate the b fragmentation uncertainty.
- ML reweighting (NEW): there is also detailed documentation about ML reweighting methods for the estimation of the b fragmentation effect that can be found in the ML reweighting twiki.

---+++ Parton shower: *Decay tables*

Last update: Feb 2024

The semi-leptonic branching ratio of B hadrons may change significantly the b jet energy response. In order to avoid generating additional samples with variations of the decay tables, we recommend reweighting the

population of b jets with semileptonic B decays at generator level. The weights are found by comparing the Pythia BRs with the measured BRs listed in the PDG and their uncertainties. The differences found and an example code of how to apply them (or to make them available in the EDM event via value maps to be applied to the gen jets produced by the PseudoTopProducer) can be found in gitlab [↗](#).

- **Correlation across the years:** this uncertainty is considered correlated for the same process across the years.
- **Correlation between singletop and ttbar:** this uncertainty is considered correlated among QCD-induced (ttbar, ttV) and EWK-induced (single-top, tV) processes.

Underlying event

Last update: Feb 2024

The differences between the nominal CP5 tune and the up/down variations of the CP5 tune in the dedicated MC samples should be quoted as the UE uncertainty.

- The CP5 tune and the up and down variations are described in TOP-16-021 [↗](#) and Eur. Phys. J. C 80 (2020) 4 [↗](#) (GEN-17-001). The exact settings for the tunes can be found in Table 7, page 36 of the paper, but can also be seen here:
 - ◆ Nominal: link to python script [↗](#)
 - ◆ Up: link to python script [↗](#)
 - ◆ Down: link to python script [↗](#)
- Correlation across the years: this uncertainty is considered correlated for the same process across the years.

Correlation between singletop and ttbar: this uncertainty is considered correlated among QCD-induced (ttbar, ttV) and EWK-induced (single-top, tV) processes.

Top pt reweighting

Last update: 6 July 2017

We recommend to apply the recipe as generally discussed in TopPtReweight

Mass of the top quark

Last update: Jul 2024

Many top quark properties, including also cross sections, are affected by the top quark mass, m_{top} . On the other hand, m_{top} is still being measured in different phase spaces with different experiments and there are differences in central values that some times are not covered by the uncertainties. In order to keep the measurements useful for future (where m_{top} converges to new values), we recommend to **not** include m_{top} as systematic but rather provide the dependence of your measurement to this SM parameter. This is possible using top quark signal samples produced with different m_{top} values.

- **Note:** in cases where providing dependence is not possible (for example for practical reasons as in differential cross sections), variations in m_{top} can be considered as systematic uncertainty.
 - ◆ The value of the uncertainty is recommended to be the latest experimental precision on this parameter (07/2024: 500 MeV, see link [↗](#)).
 - ◆ As the latest experimental precision is smaller than or at the edge of differences in m_{top} measurements at the LHC (see e.g. this thread [↗](#) in the review of TOP-17-023), we

recommend cross-checking results with inflated 1-GeV variation, reporting the average and the maximum change in the results with this uncertainty inflation.

Recommendations for single top analyses

Last update: Feb 2024

In general, all systematic uncertainties listed above have to be considered for the EWK-induced production processes as well (e.g. single top). However, there are a few exceptions that are given in this section.

- **The top pT reweighting** should ONLY be applied to top quarks from ttbar production, following the recommendations above. The top pT-correction has been derived from differential ttbar measurements. Differential measurements in single-top has not yet shown such mis-modeling (see e.g. [here](#)). Therefore the weights are not applicable for single top quark production.
- **At NLO tW production interferes with ttbar.** Two different schemes are used to remove the overlap: diagram removal (DR) and diagram subtraction (DS).
In the DR scheme all doubly resonant diagrams are removed from the tW definition. This is the default scheme used in Powheg. In the DS approach, the differential cross-section is modified by a gauge-invariant term. Besides these two global classes, MG5_aMC@NLO provides additional variations of the schemes. Please see this [link](#) for detailed definitions.
Dedicated MC samples are provided to study the different schemes (Powheg: DR/DS, MG5_aMC@NLO istr1,...,istr6 upon availability). If large differences between the schemes are observed and cannot be ruled out from additional constraints, the largest variation should be considered as systematic uncertainty. If the variations are large, differential cross sections might be provided as a function of two or more schemes.
Similar care should be taken for possible analyses of tZW, tHW, etc.
- **The hdamp variation** in Powheg single top samples is not advised to be included as uncertainty for now. Please see the dedicated discussion above.
- **Generators:** similar to ttbar here, comparisons of ME generators, e.g., Powheg vs. MG5_aMC@NLO as well as PS/hadronization models, e.g., Pythia vs. Herwig, are requested to be documented. In addition, the comparison of initial-state modeling (4FS vs. 5FS) should be made for internal documentation and review. Differential measurements are in particular asked for these studies.
- **Correlations:** for correlations across the years or between ttbar and single-top, please see the recommendations in ttbar sections.

This topic: CMS > TopSystematics

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